Fanjiang Ye

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Rice University
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Research Overview

I build efficient and scalable systems for generative AI and LLMs:

- **Diffusion / LLM / agentic inference**: caching, sparsity, and multi-level parallelism.
- **Distributed LLM training**: efficient gradient reduction and communication compression for LLMs.
- **Diffusion / LLM serving**: elastic auto parallelism and efficient scheduling design.

Education

Ph.D.	Rice University
CS	Advisor: Prof. Yuke Wang
2025–Present	Department of Computer Science
M.S.	Indiana University Bloomington
2023–2025	Advisor: Prof. Dingwen Tao
B.S.	University of Science and Technology of China
2019–2023	Advisor: Prof. Changling Zou

Employment

Applied Scientist Intern	Amazon Annapurna Labs
Summer 2026	Working on LLM Serverless Serving
Research Assistant	Rice University Yuke Lab
2025–Present	Working on Efficient ML Systems
Research Assistant	Indiana University HiPDAC Lab
2023–2025	Working on High-performance Compression

Selected Publications

* equal contribution.

S1 **Fanjiang Ye**, Zhangke Li, Xinrui Zhong, Ethan Ma, Russell Chen, Kaijian Wang, Jingwei Zuo, Desen Sun, Ye Cao, Triston Cao, Myungjin Lee, Arvind Krishnamurthy, Yuke Wang. GENSERVE: Efficient Co-Serving of Heterogeneous Diffusion Model Workloads. *arXiv preprint*, 2026.

S2 **Fanjiang Ye**, Zepeng Zhao, Yi Mu, Jucheng Shen, Renjie Li, Kaijian Wang, Saurabh Agarwal, Myungjin Lee, Triston Cao, Aditya Akella, Arvind Krishnamurthy, T. S. Eugene Ng, Zhengzhong Tu, Yuke Wang. SUPERGEN: An Efficient Ultra-high-resolution Video Generation System with Sketching and Tiling. *arXiv preprint*, 2025.

S3 Jinda Jia, Cong Xie, **Fanjiang Ye**, Hao Feng, Hanlin Lu, Daoce Wang, Haibin Lin, Zhi Zhang, Xin Liu. DUO: No Compromise to Accuracy Degradation. In *Neural Information Processing Systems (NeurIPS)*. 2025.

S4 Xiyuan Wei, Ming Lin, **Fanjiang Ye**, Fengguang Song, Liangliang Cao, My T. Thai, Tianbao Yang. Model Steering: Learning with a Reference Model Improves Generalization Bounds and Scaling Laws. In *International Conference on Machine Learning (ICML)* Spotlight. 2025.

S5 Boyuan Zhang, Bo Fang, **Fanjiang Ye**, Luanzheng Guo, Fengguang Song, Tallent Nathan, Dingwen Tao. BMQSim: Overcoming Memory Constraints in Quantum Circuit Simulation with a High-Fidelity Compression Framework. In *International Conference on Supercomputing (ICS)* Best Paper Runner-up. 2025.

S6 Boyuan Zhang, Luanzheng Guo, Jiannan Tian, Jinyang Liu, Daoce Wang, **Fanjiang Ye**, Chengming Zhang, Jan Strube, Nathan R. Tallent, Dingwen Tao. High-performance Visual Semantics Compression for AI-Driven Science. In *Principles and Practice of Parallel Programming (PPoPP)* Poster. 2025.

S7 Hao Feng, Boyuan Zhang, **Fanjiang Ye**, Min Si, Ching-Hsiang Chu, Jiannan Tian, Chunxing Yin, Zhaoxia (Summer) Deng, Yuchen Hao, Pavan Balaji, Tong Geng, Dingwen Tao. Accelerating Communication in Deep Learning Recommendation Model Training with Dual-Level Adaptive Lossy Compression. In *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*. 2024.

Other Publications

O1 Jingwei Zuo, Xinze Feng, Zien Liu, Kaijian Wang, **Fanjiang Ye**, Ye Cao, Zhuang Wang, Yuke Wang. ALTO: Adaptive LoRA Tuning and Orchestration for Heterogeneous LoRA Training Workloads. *arXiv preprint*, 2026.

O2 Yihua Liu, **Fanjiang Ye**, Bowen Lin, Rongyu Fang, Chengming Zhang. TIDE: Text-Informed Dynamic Extrapolation with Step-Aware Temperature Control for Diffusion Transformers. *arXiv preprint*, 2025.

O3 Xinheng Ding, Tianyi Zhang, Hao Li, **Fanjiang Ye**, Wenya Xie, Qingquan Song, Yuke Wang, Zirui Liu. Teach to Fish, Not to Feed: Internalizing Retrieval for External Memory in Large Language Models. *arXiv preprint*, 2025.

O4 Xinrui Zhong, Xinze Feng, Jingwei Zuo, **Fanjiang Ye**, Yi Mu, Junfeng Guo, Heng Huang, Myungjin Lee, Yuke Wang. An Efficient and Adaptive Watermark Detection System with Tile-based Error Correction. *arXiv preprint*, 2025.

O5 Bowen Lin, **Fanjiang Ye**, Yihua Liu, Zhenghui Guo, Boyuan Zhang, Weijian Zheng, Yufan Xu, Tiancheng Xing, Yuke Wang, Chengming Zhang. SDiT: Semantic Region-Adaptive for Diffusion Transformers. *arXiv preprint*, 2025.

O6 Xiyuan Wei, **Fanjiang Ye**, Ori Yonay, Xingyu Chen, Dingwen Tao, Tianbao Yang. FastCLIP: A Suite of Optimization Techniques to Accelerate CLIP Training with Limited Resources. *arXiv preprint*, 2024.

Awards and Honors

- Indiana University Travel Awards 2024
- USTC Fellowship 2023
- USTC Outstanding Student Scholarship 2020, 2021, 2022

Professional Service

Artifact Evaluation Committee

- ACM Conference on AI and Agentic Systems (CAIS) 2026
- Conference on Machine Learning and Systems (MLSys) 2026
- European Conference on Computer Systems (EuroSys) Fall 2026
- ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) Summer 2026
- ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP) 2026
- ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) Spring 2026
- ACM Symposium on Operating Systems Principles (SOSP) 2025

Conference Program Committee

- Neural Information Processing Systems (NeurIPS) 2026
- European Conference on Computer Vision (ECCV) 2026
- Computer Vision and Pattern Recognition (CVPR) 2026
- IEEE International Conference on Quantum Computing and Engineering (QCE) 2024

Teaching Experience

- COMP 422/534: Parallel Computing Rice University Spring 2026
- ENGR-E 516: Cloud Computing Indiana University Spring 2025